

SAMBIT MISHRA

PhD Student, Electrical & Computer Engineering | University of Southern California

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Research Focus

Causal inference and probabilistic graphical models, with emphasis on identifiability theory, design of efficient discovery algorithms for identifiable causal DAGs, and analysis on bounds and design of lowest variance discovery algorithms for non-identifiable or partially-identifiable cases, with focus on scalable continuous optimization methods for causal discovery in high-dimensional data, and mixed distribution-based datasets.

Education

University of Southern California (USC)

Ph.D. in Electrical and Computer Engineering; Advisor: Prof. Urbashi Mitra

GPA: 4.00/4.00

Key Coursework: Probability Theory (EE 503 - Fall 2025, Grade: A), Linear Algebra (EE 510 - Fall 2025, Grade: A), Supervised Machine Learning (EE 559 - Spring 2026, Grade: A), Inference & Estimation Theory (EE 563 - Spring 2026, Grade: A).

Los Angeles, CA, USA

Aug 2025 – Present

Indian Institute of Technology (IIT) Bhubaneswar

B.Tech. (Hons.) in Electronics and Communication Engineering; Advisor: Dr. Soumya P. Dash

GPA: 9.14/10.00

Bhubaneswar, Odisha, India

2021 – 2025

Research Experience

Graduate Research Assistant

Ming Hsieh Department of Electrical and Computer Engineering, USC

Los Angeles, CA, USA

Aug 2025 – Present

- Developing continuous optimization methods for scalable DAG estimation using differentiable constraints to reduce iterations and enable large-scale batching.
- Designed a soft-intervention selection framework to improve causal discovery efficiency and lower experimental costs.
- Studying identifiability of causal graphs under finite samples and latent variable settings.
- Built GPU-accelerated structure learning pipelines in PyTorch using Adam and L-BFGS for non-convex optimization.

Undergraduate Researcher

School of Electrical and Computer Sciences, IIT Bhubaneswar

Bhubaneswar, Odisha, India

Aug 2024 – May 2025

- Developed SER-optimized modulation schemes for RIS-assisted non-coherent communication systems.
- Conducted error analysis for multi-level ASK modulations, optimizing energy efficiency.

Publications

- [1] C. Peng, **S. Mishra**, U. Mitra, “Learning to Intervene: Optimized Soft Intervention Selection for Causal Discovery,” *IEEE Int. Conf. on Acoustics, Speech and Signal Processing (ICASSP)*, accepted, 2026.
- [2] **S. Mishra**, S. P. Dash and G. C. Alexandropoulos, “SER-Optimized Multi-Level ASK Modulations for RIS-Assisted Communications With Energy- and Sign-Based Noncoherent Reception,” in *IEEE Transactions on Green Communications and Networking*, vol. 10, pp. 1433-1445, 2026, doi: 10.1109/TGCN.2025.3633182.
- [3] **S. Mishra** and S. P. Dash, “Error Analysis With Optimal Receiver and Multi-Level ASK for RIS-Assisted Noncoherent Wireless System,” in *IEEE Wireless Communications Letters*, vol. 15, pp. 300-304, 2026, doi: 10.1109/LWC.2025.3624154.

Awards & Academic Distinctions

- Secured All India Rank 4000 among ~1,000,000 candidates in JEE Advanced 2021, India
- Awarded National Talent Search Scholarship by achieving a rank among the top 2000 students among ~1,200,000 candidates in National Talent Search Examination (NTSE) 2019, Government of India
- Qualified Regional Mathematical Olympiad (RMO) 2019; Selected participant, Indian National Mathematical Olympiad (INMO) 2020

Technical Skills

- Causal Inference:** Causal Discovery, Bayesian Networks, Structural Equation Models
- Optimization:** Gradient-based solvers (Adam, RMSProp, L-BFGS), Convex Optimization, Bilevel Optimization.
- Programming & Tools:** Python (PyTorch, NumPy, SciPy), C++, CUDA, MATLAB, Git, LaTeX.

Industry Experience

NVIDIA Corporation

ASIC Engineering Intern, CHI VIP Team

Bengaluru, Karnataka, India

May 2024 – July 2024

Verified CHI protocol compliance for CPU/GPU IPs using SystemVerilog and Perl automation; ensured interoperability between internal and external verification IPs.